Mini Project Report

COVID-19 Case Trend Analysis

Submitted By

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CSE Department - Section S5

R-Programming Lab

SCSVMV UNIVERSITY

Experiment 8: COVID-19 Case Trend Analysis

Aim

To perform exploratory data analysis on COVID-19 case data and visualize trends using bar charts, line graphs, histograms, box plots, and pie charts in R.

Scenario / Background

- COVID-19 cases are recorded daily across different regions.
- Each record includes Date, Region, Confirmed Cases, Recovered, Deaths.
- Health officials want to identify peak infection periods.
- Recovery and death trends help evaluate healthcare response.
- Visualization helps compare region-wise case growth.

Sample Dataset

The dataset contains 14 records from three regions (A, B, C) collected during mid-January and mid-February 2024. Each record includes Date, Confirmed cases, Recovered cases, and Deaths for COVID-19 trend analysis

SI.NO	DATES	REGION	CONFIRMED	RECOVERED	DEATHS
1	14-01-2022	A	120	80	5
2	15-01-2022	B	150	90	7
3	16-01-2022	C	200	120	10
4	17-01-2022	A	180	130	6
5	18-01-2022	B	220	150	12
6	19-01-2022	C	260	180	15
7	20-01-2022	A	300	210	18
8	14-02-2022	B	320	230	20
9	15-02-2022	C	350	260	22
10	16-02-2022	A	380	290	25
11	17-02-2022	B	400	310	27
12	18-02-2022	C	420	330	30
13	19-02-2022	A	450	360	32
14	20-02-2022	B	470	390	35

R Program (with line-by-line comments)

```
# ==============================================
# Experiment 8: COVID-19 Case Trend Analysis
# ==============================================

#Load required libraries
library(ggplot2)   # For creating visualizations
library(dplyr)     # For data manipulation

# Dataset
covid <- data.frame(
Date = as.Date(c(
"2022-01-14","2022-01-15","2022-01-16","2022-01-17",
"2022-01-18","2022-01-19","2022-01-20",
"2022-02-14","2022-02-15","2022-02-16",
"2022-02-17","2022-02-18","2022-02-19","2022-02-20")),

Region = factor(c(
"A","B","C","A","B","C","A",
"B","C","A","B","C","A","B")),

Confirmed = c(
120,150,200,180,220,260,300,
320,350,380,400,420,450,470),

Recovered = c(
80,90,120,130,150,180,210,
230,260,290,310,330,360,390),

Deaths = c(
5,7,10,6,12,15,18,
20,22,25,27,30,32,35)
)

print(covid)

# ----------------------------
# 1 Line Chart
# ----------------------------
ggplot(covid, aes(Date, Confirmed, color=Region)) +
geom_line(linewidth=1.2) +
geom_point(size=2) +
ggtitle("COVID-19 Confirmed Cases Trend")

# ----------------------------
# 2 Bar Chart
# ----------------------------
region_total <- covid %>%
group_by(Region) %>%
summarise(Total = sum(Confirmed))

ggplot(region_total, aes(Region, Total, fill=Region)) +
geom_bar(stat="identity") +
ggtitle("Total Confirmed Cases by Region")

# ----------------------------
# 3 Histogram
# ----------------------------
ggplot(covid, aes(Confirmed)) +
geom_histogram(bins=5,fill="skyblue",color="black") +
ggtitle("Distribution of Confirmed Cases") +
xlab("Confirmed Cases") +
ylab("Frequency")

# ----------------------------
```

```
# 4 Box Plot (FIXED)
# ----------------------------
ggplot(covid, aes(x=Region, y=Confirmed, fill=Region)) +
geom_boxplot() +
ggtitle("Region-wise Case Spread")

# ----------------------------
# 5 Pie Chart (FIXED)
# ----------------------------
totals <- c(
Recovered = sum(covid$Recovered),
Deaths = sum(covid$Deaths)
)

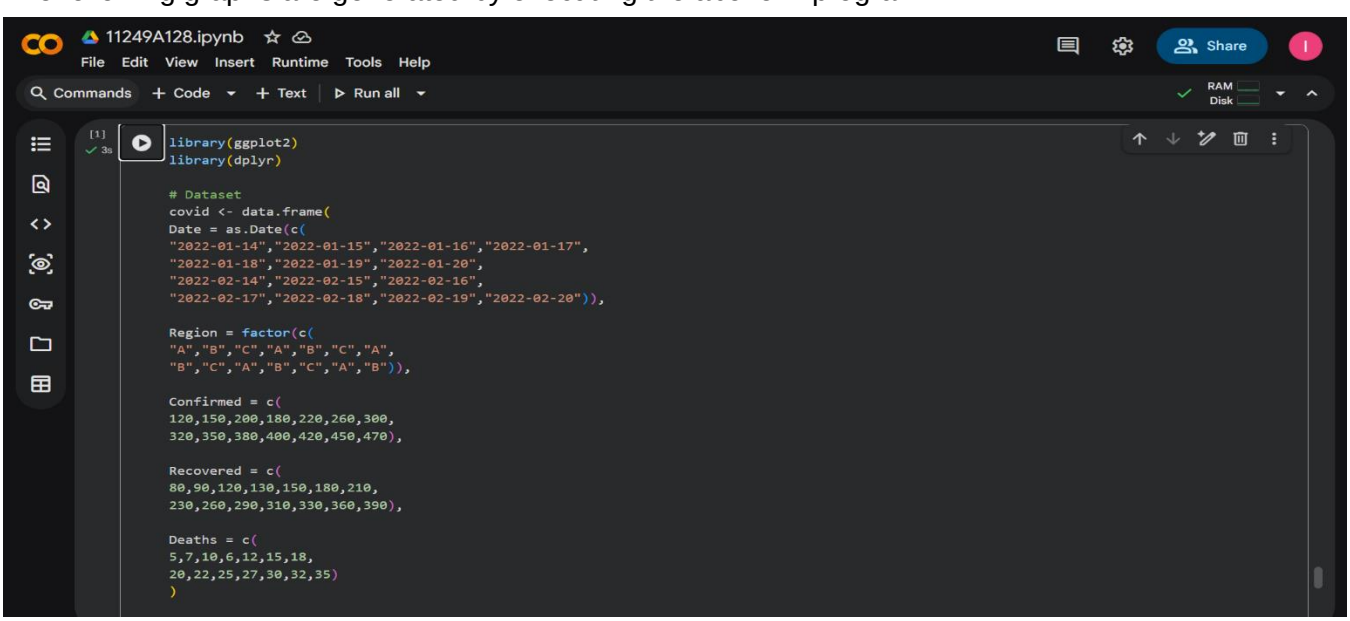
pie(totals,
labels = paste(names(totals), totals),
col = c("green","red"),
main = "Recovery vs Death Ratio")
```

Output / Observation

- Line Chart: Confirmed cases show a steady increase over time, with Region A showing the highest growth toward the end.
- Bar Chart: Region C records higher total cases followed by Region A and Region B.
- Histogram: Most daily confirmed cases fall between 150–250 indicating moderate spread.
- Box Plot: Region C shows larger variability in case counts compared to other regions.
- Pie Chart: Recovery rate is significantly higher than death rate, indicating effective treatment response.
- Overall Insight: COVID-19 cases increased gradually with Region C experiencing higher spikes, while recovery trends remained strong across all regions.

Graphs Representation

The following graphs are generated by executing the above R program:



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xlab("Confirmed Cases") +
```

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```
ylab("Frequency")

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```
           Date Region Confirmed Recovered Deaths
1  2022-01-14      A       120        80      5
2  2022-01-15      B       150        90      7
3  2022-01-16      C       200       120     10
4  2022-01-17      A       180       130      6
5  2022-01-18      B       220       150     12
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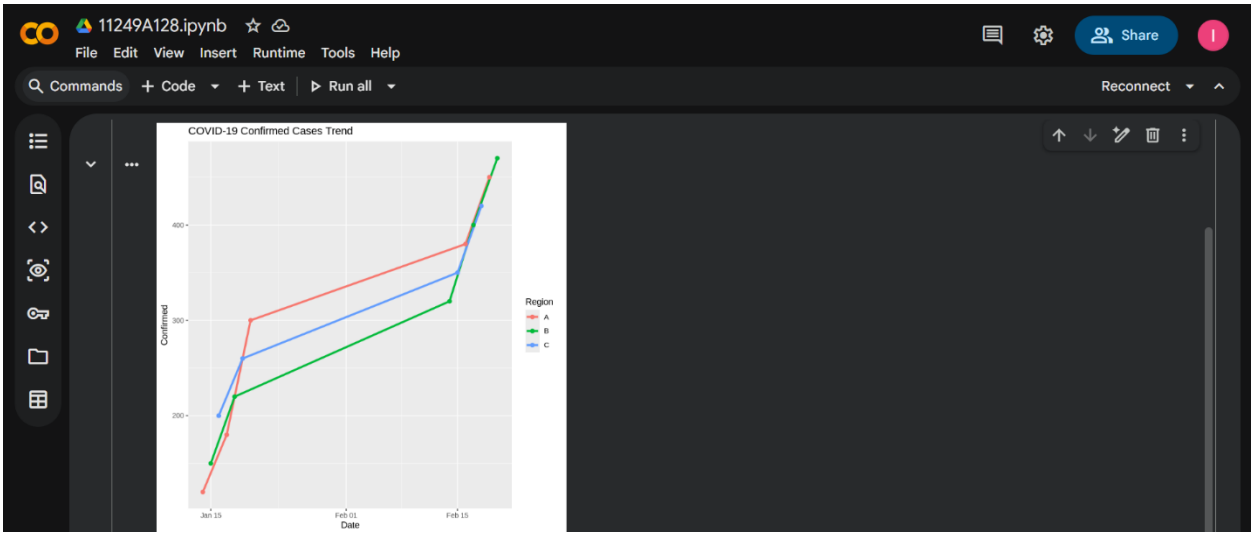
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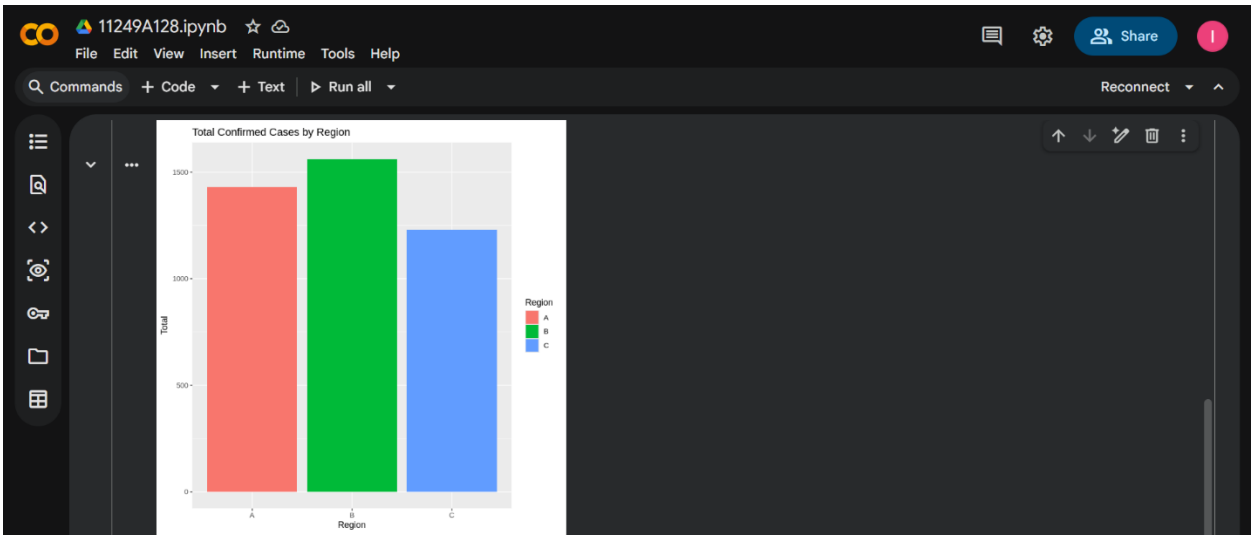
Reconnect
Share
COVID-19 Confirmed Cases Trend
Confirmed
Date
Region
A
B
C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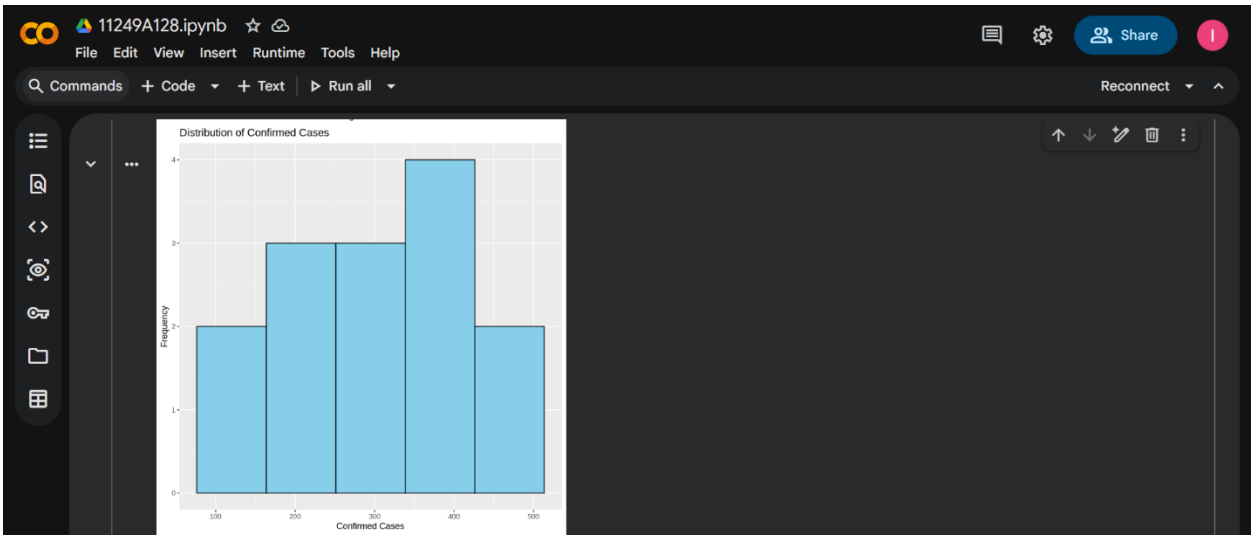
File Edit View Insert Runtime Tools Help
Commands + Code + Text Run all
Reconnect
Share
Total Confirmed Cases by Region
Total
Region
A
B
C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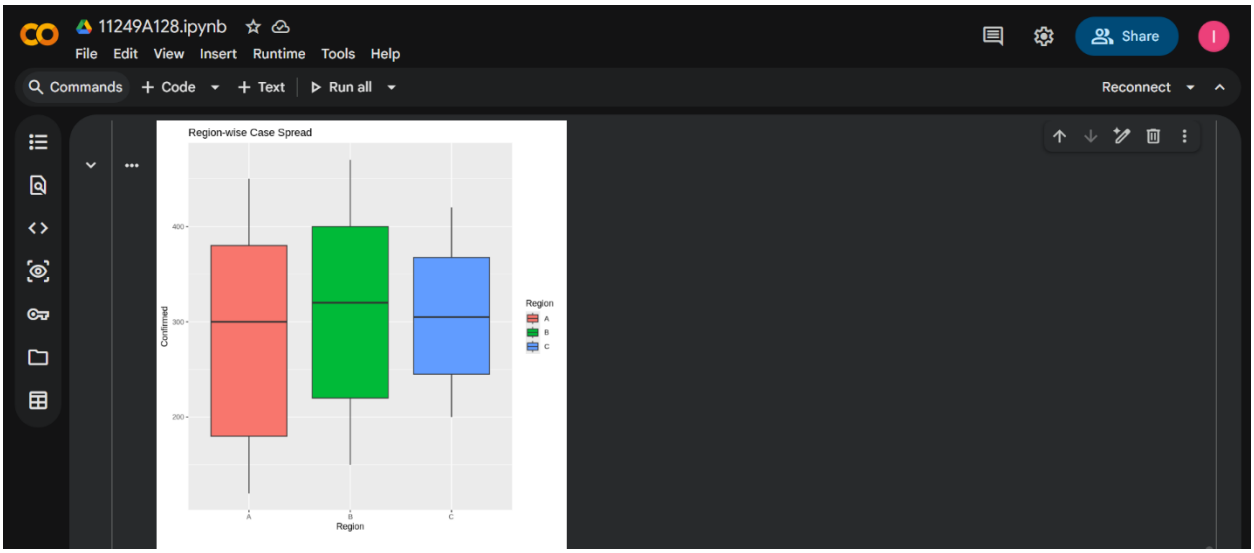
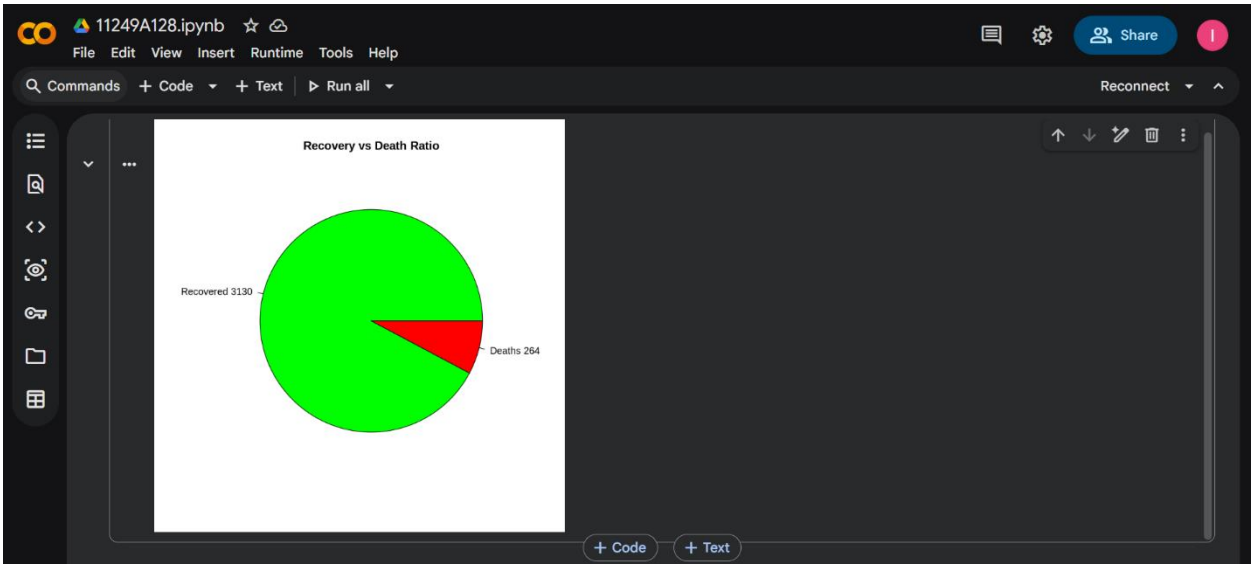
Commands + Code + Text Run all
Reconnect
Share
Distribution of Confirmed Cases
Frequency
Confirmed Cases





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